

2022 Trial Higher School Certificate PRELIMINARY EXAMINATION

Mathematics Standard

- **Solutions including marking scale**
- **Mapping grid**

We have endeavoured to ensure that the solutions are free of errors and follow the spirit of the syllabus in the methods used to solve the problems. However, individual teachers may opt for alternate solutions and/or may select a different marking system.

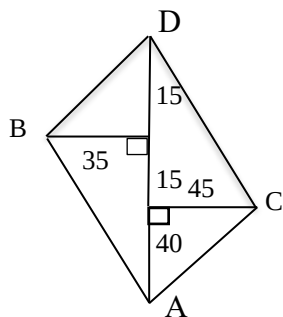
Marking Guidelines: Mathematics Standard, Preliminary HSC Trial Examination

1	C
2	D
3	B
4	B
5	A
6	C
7	C
8	D
9	A
10	D
11	C
12	D
13	B
14	C
15	B

ANSWERS

Question 16 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1



Question 16 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$AB^2 = 55^2 + 35^2 \text{ i.e. } AB = 65.2$$

$$BD^2 = 35^2 + 15^2 \text{ i.e. } BD = 38.1$$

$$DC^2 = 30^2 + 45^2 \text{ i.e. } DC = 54.1$$

$$AC^2 = 45^2 + 40^2 \text{ i.e. } AC = 60.2$$

$$\text{Perimeter} = 217.6 \text{ i.e. } 218 \text{ m}$$

Question 16 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$A = \frac{1}{2}(70)35 + \frac{1}{2}(70)45 = 1225 + 1575 = 2800m^2$$

Question 17 (a)

Criteria	Marks
• Provides correct solution	1

$$\text{Medicare Levy} = 0.26(23\,490 - 19\,500) = \$1\,037.40$$

Question 17 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\text{Liong's Medicare Levy} = 4.75/100 \times 38\,438 = \$1\,825.81$$

$$\text{Difference in Medicare Levy} = \$1\,825.81 - 1\,037.40 = \$788.41$$

Question 18 (a)

Criteria	Marks
• Provides correct solution	1

Dimitri's weekly wage

$$= 5(15.45) + 3.5 \times 15.45 + 4.5 \times 1.5 \times 15.45 = \$235.61$$

Question 18 (b)

Criteria	Marks
• Provides correct solution	1

$$\text{Total cost} = 935 + 48 + 75 + 2400 = \$3\,458$$

Question 18 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\text{Savings per week} = 40/100 \times \$235.61 = \$94.24$$

$$\text{Number of weeks} = 3\,458/94.24 = 36.69 \text{ weeks i.e. } 37 \text{ weeks}$$

Question 19 (a)

Criteria	Marks
• Provides correct solution	1

$$\text{Number of girls} = 76 + 20 + 10 + 5 + 2 = 113$$

Question 19 (b)

Criteria	Marks
Provides correct solution	1

P(at least one girl visited the shopping centre during the week) = 37/113

Question 20 (a)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

Area of rectangle = $35(34) = 1190 \text{ m}^2$
 Volume of rainfall = $1190 \text{ m}^2 \times 84 \text{ mm}$
 $= 1190(0.084) \text{ m}^3$
 $= 99.96 \text{ m}^3$
 $= 99.96 \text{ kL}$
 $= 100 \text{ kL}$

Question 20 (b)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$V = \pi r^2 h$
 $100 = \pi(3/2)^2 h$
 $h = 14.15 \text{ m}$

Question 20 (c)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

Volume of empty tank = $\pi(3/2)^2 18 = 127.24 \text{ m}^3$
 Difference in volume = $127.24 - 100 = 27.24 \text{ kL}$
 i.e. 27.24 kL of rain needs to fall before the tank overflows.

Question 21

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$: C = \sqrt{5s^2}$$

$$C^2 = 5s^2$$

$$s^2 = \frac{C^2}{5}$$

$$s = \sqrt{\frac{C^2}{5}}$$

Question 22

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$p(p-q) - q(q-p) = p^2 - pq - q^2 + pq = p^2 - q^2$$

Question 23 (a)

Criteria	Marks
• Provides correct solution	1

$$C = \frac{nA}{n+12}$$

$$C = \frac{5(28)}{5+12} = 8.24mL$$

Question 23 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$C = \frac{nA}{n+12}$$

$$C = \frac{9(28)}{9+12} = 12mL$$

$$\text{Difference is } 12 - 8.24 = 3.76 \text{ mL}$$

Question 24 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Female

9	6	5	2
4	3	6	1
	6	7	3
4	0	8	4
8	3	1	9

Male

Question 24 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Male pulse rate: Mean = 76.4 and standard deviation = 14.42

Female pulse rate: Mean = 77.1 and standard deviation = 13.71

Question 24 (c)

Criteria	Marks
• Provides correct solution	3
• Majority of solution is correct	2
• Provides some relevant information	1

Male

Lower extreme = 56

 $Q_L = 63$

Median = 78

 $Q_R = 91$

Upper extreme = 98

Female

Lower extreme = 52

 $Q_L = 68$

Median = 77

 $Q_R = 86$

Upper extreme = 99

Question 25 (a)

Criteria	Marks
Provides correct solution	1

$$V = \frac{1}{2}(2+5)(4 \times 3) = 42m^3$$

Question 25 (b)

Criteria	Marks
Provides correct solution	1

For one container, capacity = 42 kL

For 20 containers, capacity is $42 \times 20 = 840$ kL

Question 25 (c)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

Absolute error = $\pm 0.5m$

Limits of accuracy = 3.5 to 4.5 m

Question 25 (d)

Criteria	Marks
Provides correct solution	2
Provides some relevant information	1

$$\text{Length percentage error} = \frac{\pm 0.5}{4} \times 100\% = \pm 12.5\%$$

$$\text{Height Percentage error} = \frac{\pm 0.5}{3} \times 100\% = \pm 16.7\%$$

Question 25 (e)

Criteria	Marks
Provides correct solution	1

The height measurement as it has a smaller percentage error.

Question 26 (a)

Criteria	Marks
Provides correct solution	1

$$\text{Cost} = 4000(0.0768) + 3000(0.0322) = \$403.80$$

Question 26 (b)

Criteria	Marks
• Provides correct solution	3
• Majority of solution is correct	2
• Provides some relevant information	1

Usage = $7000 \times 102.5/100 = 7175$ MJ

i.e for a 3 month period, usage is 7175 MJ

Cost over 3 month period = $4000(0.0768) + 3175(0.0322) = \409.44

Over a 12 month period, cost is $4(409.44) = \$1\,637.76$

Question 27 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$A = \frac{1}{2}(2.1+1.2)(1.9) + \frac{1}{2}(1.2+1.9)(1.9) + \frac{1}{2}(1.9+2.2)(1.9) + \frac{1}{2}(2.2+1.8)(1.9)$$

$$A = 13.775$$

$$A = 13.8\text{m}^2$$

Question 27 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Area of tile is 30 cm x 15 cm i.e. $(0.3 \text{ m} \times 0.15 \text{ m} = 0.045 \text{ m}^2)$

Each box covers 1.125 m^2 (i.e $0.045 \times 25 = 1.125 \text{ m}^2$)

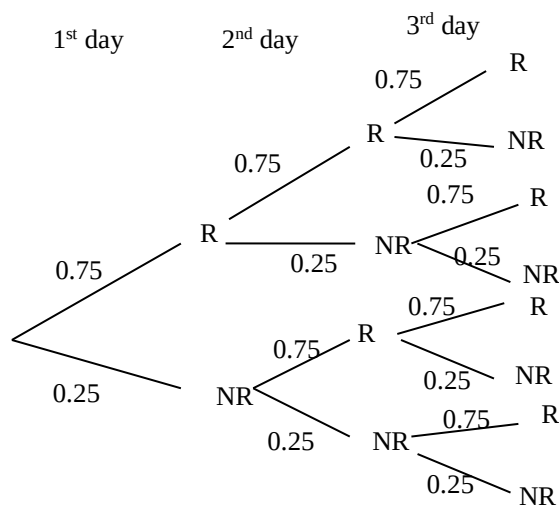
Total number of boxes = $13.8/1.125 = 12.2666..$

i.e 13 boxes are needed

Cost to purchase 13 boxes = $13(\$76.65) = \996.45

Question 28 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1



Question 28 (b)

Criteria	Marks
• Provides correct solution	1

$$P(\text{rain falling on all three nights}) = 0.75 \times 0.75 \times 0.75 = 0.421875 = 0.42$$

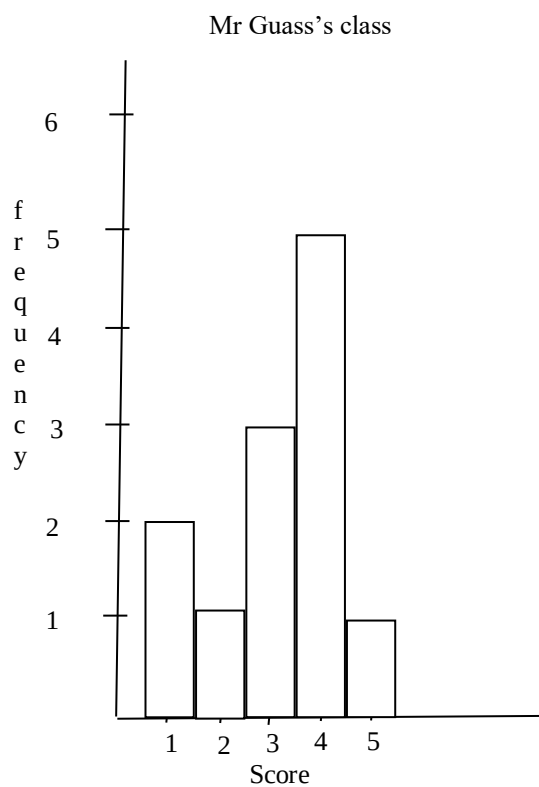
Question 28(c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$\begin{aligned}
 P(\text{rain falling on at least one night}) &= 1 - P(\text{NR}) \\
 &= 1 - (0.25 \times 0.25 \times 0.25) \\
 &= 0.9844 \\
 &= 0.98
 \end{aligned}$$

Question 29 (a)

Criteria	Marks
• Provides correct solution	3
• Majority of solution is correct	2
• Provides some relevant information	1

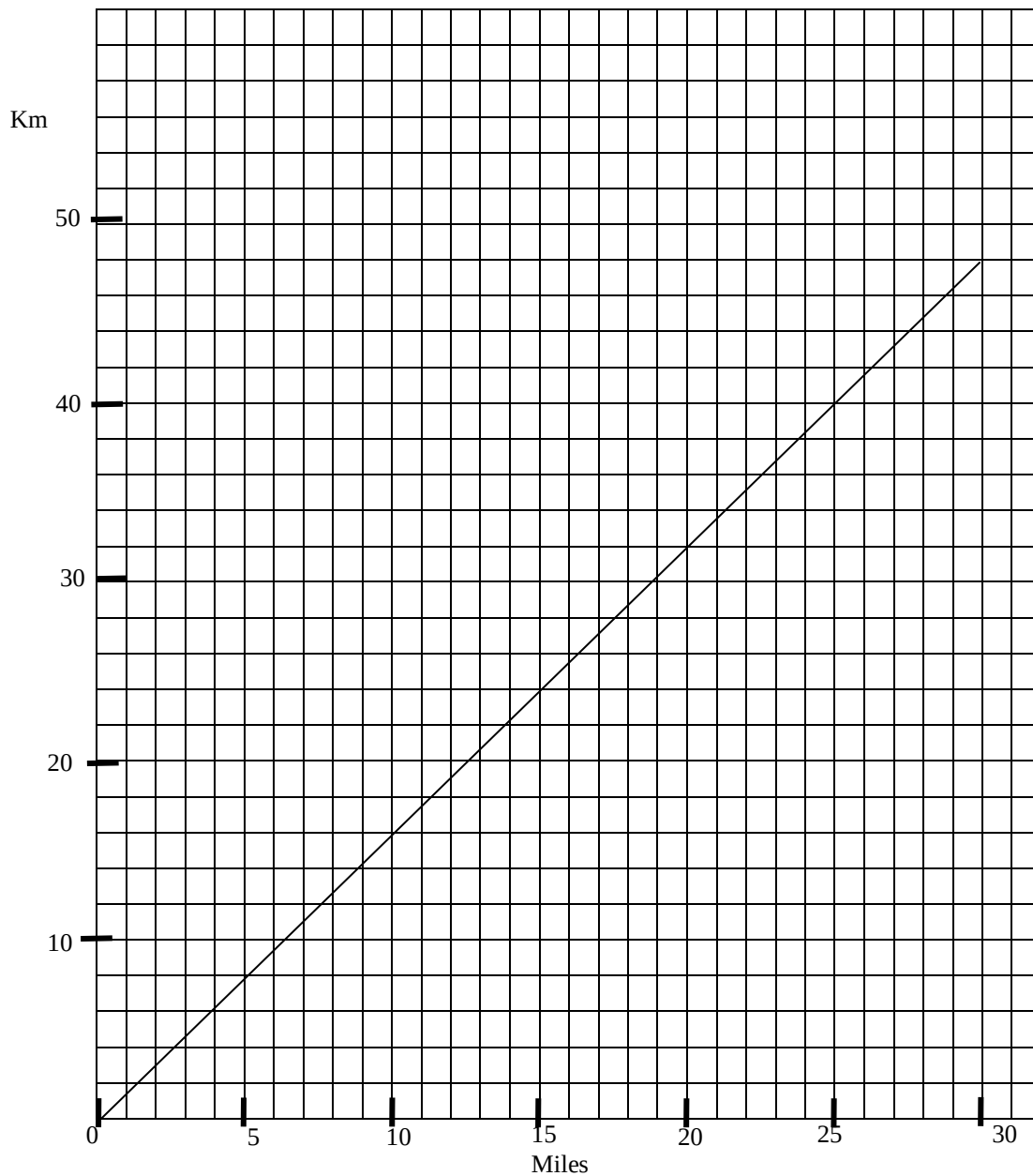
**Question 29 (b)**

Criteria	Marks
• Provides correct solution	1

$$\text{Mean} = 38/12 = 3\frac{1}{6}$$

Question 30 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

**Question 30 (b)**

Criteria	Marks
• Provides correct solution	1

$$K = \frac{8}{5}m$$

Question 30 (c)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Gradient = $\frac{8}{5}$

It is the conversion rate in which miles is converted to kilometres i.e every 5 miles travelled is equivalent to 8 km.

Question 30 (d)

Criteria	Marks
• Provides correct solution	1

18 miles = $\frac{8}{5}(18) = 28.8$ km

Question 31 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Let T be number of hours before Leila can drive.

$$0.193 - 0.019(T) = 0.05$$

$$T = (0.193 - 0.05) / 0.019$$

$$T = 7.526$$

i.e. 7 hours and 32 minutes

Question 31 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

$$0.193 - 0.019(T) = 0$$

$$T = 0.193 / 0.019 = 10.157$$

$$T = 10 \text{ hours } 9 \text{ mins}$$

Leila's BAC will reach zero after 10 hours and 9 minutes.

Question 32 (a)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Left Sydney 10 am Saturday

Time difference between Sydney and El Salvador = $10 + 6 = 16$ hours

i.e left at 6pm Friday El Salvador time (10am Sat Sydney time – 16 hrs)

Question 32 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Total flight time (including time in Iran) = 9hr 15 mins + 22hr 4mins + 11 hr 30 mins
i.e. 1 day 18 hrs and 49 mins

Plane lands in El Salvador (6pm Friday + 1 day 18 hrs 49 mins) = 12:49 pm Sunday

Question 33 (a)

Criteria	Marks
• Provides correct solution	1

Final balance = \$700 x 1.1025 = \$771.75

Question 33 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

4% p.a = 2% half yearly and 3 years = 6 periods

Final balance = 600 x 1.1262 = \$675.72

Compound interest = \$675.72 – 600 = \$75.72

Question 34 (a)

Criteria	Marks
• Provides correct solution	4
• Majority of solution is correct	3
• Provides correct surface area of inner and outer cylinders	2
• Provides some relevant information	1

Surface area of outer cylinder = $2\pi(35)80 = 17592.92\text{cm}^2$

Surface area of inner cylinder = $2\pi(32)80 = 16084.95\text{cm}^2$

Surface area of top of container = $\pi(35)^2 - \pi(32)^2 = 631.46\text{cm}^2$

Surface area of top of table = $70 \times 90 - \pi \times 35 \times 35 = 2451.55\text{cm}^2$

Surface area of base of table = $70 \times 90 - 4 \times 7 \times 7 = 6104\text{cm}^2$

Surface area of sides of table = $2 \times 70 \times 12 + 2 \times 90 \times 12 = 3840\text{cm}^2$

Surface area of legs = $4 \times [2(7 \times 50 + 7 \times 50) + 49] = 5796\text{cm}^2$

Total surface area = 52 500.88 cm^2

Question 34 (b)

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Total surface area = $(52\,500.88\text{cm}^2 \times 300)/10000\text{ m}^2 = 1\,575.03\text{m}^2$

Each tin of varnish covers 60 m^2 . Thus $1\,575.03/60 = 26.25$ cans

i.e. 27 cans are needed at a cost of $\$29.50 \times 27 = \796.50

Question 35

Criteria	Marks
• Provides correct solution	2
• Provides some relevant information	1

Let A be the taxable income.

$$37630 = 15\,580 + (A - \$60000) \times 0.47$$

$$37630 - 15\,589 = (A - \$60000) \times 0.47$$

$$22050 = (A - \$60000)(0.47)$$

$$46\,914.89 = A - 60\,000$$

$$A = \$106\,914.89$$

Mathematics Standard

Preliminary HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes	Targeted Performance bands
1	1	MS-A1: Formulae and equations	MS11-1,6,9,10	1-2
2	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	1-2
3	1	M1.1: Measurement and energy	MS11-3,4,9,10	2-3
4	1	MS-A2: Linear relationships	MS11-1,6,9,10	2-3
5	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	2-3
6	1	MS-S2: Relative frequency and probability	MG11-1,6,9,10	2-3
7	1	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
8	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
9	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	1-2
10	1	MS-M2: Working with time	MG11-1,6,9,10	3-4
11	1	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
12	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
13	1	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
14	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
15	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
16 (a)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
16 (b)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
16 (c)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
17 (a)	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
17 (b)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
18 (a)	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
18 (b)	1	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
18 (c)	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	2-3
19 (a)	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
19 (b)	2	MS-S2: Relative frequency and probability	MG11-1,6,9,10	3-4
20 (a)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	4-5
20 (b)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	4-5
20 (c)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	4-5
21	2	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
22	2	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
23 (a)	1	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3
23 (b)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	2-3

24 (a)	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
24 (b)	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
24 (c)	3	S1.2: Exploring and describing data	MS11-2,7,8,9,10	2-3
25 (a)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
25 (b)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
25 (c)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
25 (d)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
25 (e)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
26 (a)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
26 (b)	3	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
27 (a)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
27 (b)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	3-4
28 (a)	2	M1.2: Perimeter, area and volume	MS11-3,4,9,10	2-3
28 (b)	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	3-4
28 (c)	2	S1.2: Exploring and describing data	MS11-2,7,8,9,10	3-4
29 (a)	3	S1.2: Exploring and describing data	MS11-2,7,8,9,10	3-4
29 (b)	1	S1.2: Exploring and describing data	MS11-2,7,8,9,10	3-4
30 (a)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
30 (b)	1	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
30 (c)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
30 (d)	1	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
31 (a)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
31 (b)	2	MS-A1: Formulae and equations	MS11-1,6,9,10	3-4
32 (a)	2	MS-M2: Working with time	MG11-1,6,9,10	5-6
32 (b)	2	MS-M2: Working with time	MG11-1,6,9,10	5-6
33 (a)	1	F1.1: Interest and depreciation	MS11-2,5,6,9,10	4-5
33 (b)	2	F1.1: Interest and depreciation	MS11-2,5,6,9,10	4-5
34 (a)	1	M1.2: Perimeter, area and volume	MS11-3,4,9,10	5-6
34 (b)	4	M1.2: Perimeter, area and volume	MS11-3,4,9,10	5-6
35	2	F1.2: Earning and managing money	MS11-2,5,6,9,10	3-4